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your question will not be graded.

Your work may be submitted on your own paper, by writing on this worksheet printed out, or by writing via tablet, among possibly other options. If you are having problems submitting your work, please attend office hours. (There is no late penalty for submitting the assignment on the due date, but there will be no late submissions accepted for any reason.) Click on [Gradescope's help page](#) and read their instructions.

1. Using algebra and intuition, determine the exact value of the limits of the recursively defined sequences.

(a) $a_{n+1} = \frac{1}{2} \left(a_n + \frac{2}{a_n} \right)$ with $a_0 = 1$.

(b) $a_{n+1} = \frac{1}{2} \left(a_n + \frac{3}{a_n} \right)$ with $a_0 = 1$.

(c) $a_{n+1} = \frac{2}{3}a_n + \frac{4}{3a_n^2}$ with $a_0 = 1$

2. Let θ be a (constant but currently unknown) real number. Determine, with proper justification, if the sequence

$$\left\{ \frac{\sin(n^2\theta)}{n} \right\}_{n=1}^{\infty}$$

converges. If it does, find the value of the limit.

3. Determine whether the series

$$\sum_{n=1}^{\infty} \ln\left(\frac{n}{n+2}\right)$$

converges or diverges. If it converges, find its sum.

4. Determine whether the series $\sum_{n=1}^{\infty} \frac{n - 5\sqrt{n}}{1 + 3n^2}$ is convergent or divergent.

5. Find the value(s) of x which makes the equation true, or explain why one does not exist.

$$\frac{9}{10} = \sum_{n=2}^{\infty} \left(1 + \frac{3x}{2}\right)^{-n}$$

6. Consider dropping a basketball from a height of five feet while standing on a flat surface. If the ball rebounds from each bounce to a point 85% as high as its previous height...

- (a) ...find the total distance that the ball travels (up *and* down) before coming to rest (after an infinite number of bounces!).

- (b) ...find the total time necessary for the ball to come to rest (after an infinite number of bounces!).

Use the fact that the amount of time it takes for an object to fall from height h to the ground can be found from $h = \frac{1}{2}gt^2$. Use $g = 32.2$ ft per second per second.

7. Determine whether the series is convergent or divergent. If it is convergent, find its sum in two different ways.

$$\frac{1}{4} + \frac{2}{16} + \frac{1}{64} + \frac{2}{256} + \frac{1}{1024} + \frac{2}{4096} + \cdots$$

8. Determine, with justification, whether the series $\sum_{n=1}^{\infty} \left(\sqrt{n^2 + 1} - n \right)$ converges or diverges.

The remaining problems outline a rigorous method for determining the partial sum formula for a telescoping series (and some others).

9. Consider the series $\sum_{n=1}^{\infty} (e^n - e^{n+1})$. Write a_n for $e^n - e^{n+1}$ and S_N for $\sum_{n=1}^N a_n$.

(a) Compute exact values for S_1 , S_2 , and S_3 .

(b) Make an educated guess of the general form for S_N .

- Plug in $N = 3$ in the previous part to verify that you get the same answer as S_3 in (a). If not, go back.

(c) Based on your format above, determine what S_{N+1} should be, i.e. replace each occurrence of N by $N + 1$.

(d) Compute S_{N+1} a different way, using $S_{N+1} = S_N + a_{N+1}$.

- If the previous two steps match, then your guess for the general form for S_N was correct. If not, then go back and fix the earlier steps.

(e) Compute the limit of S_N to find the value of the series, if it converges, or conclude that it doesn't converge.

10. Consider the series $\sum_{n=2}^{\infty} \frac{2}{n^2 - 1}$. Write a_n for $\frac{2}{n^2 - 1}$ and S_N to mean the partial sum $\sum_{n=2}^N a_n$.

- (a) Rewrite a_n so that the series will be in a more suitable form.

- (b) Compute exact, partially simplified values of S_2 , S_3 , S_4 , and S_5 . Hint: cancel terms, but otherwise do not simplify.

- (c) Make an educated guess of the general form for S_N .

- Plug in $N = 5$ in the previous part to verify that you get the same answer as S_5 in (b). If not, go back.

- (d) Based on your format above, determine what S_{N+1} should be, i.e. replace each occurrence of N by $N + 1$.

- (e) Compute S_{N+1} a different way, using $S_{N+1} = S_N + a_{N+1}$.

- If the previous two steps match, then your guess for the general form for S_N was correct. If not, then go back and fix the earlier steps.

- (f) Compute the limit of S_N to find the value of the series, if it converges, or conclude that it doesn't converge.

11. Consider the series $\sum_{n=3}^{\infty} \frac{3n-4}{n^3-3n^2+2n}$. Write a_n as usual and S_N for $\sum_{n=3}^N a_n$.

(a) Rewrite a_n so that the series will be in a more suitable form.

(b) Compute exact, partially simplified values of some partial sums.

(c) Make an educated guess of the general form for S_N .

- Plug in a specific value of N in (b) to verify that you get the same answer as in (a). If not, revisit (b) and (c).

(d) Based on your format above, determine what S_{N+1} should be, i.e. replace each occurrence of N by $N+1$.

(e) Compute S_{N+1} a different way, using $S_{N+1} = S_N + a_{N+1}$.

- If the previous two steps match, then your guess for the general form for S_N was correct. If not, then go back and fix the earlier steps.

(f) Compute the limit of S_N to find the value of the series, if it converges, or conclude that it doesn't converge.